

EX:NO:11**PROGRAM CODE:**

```
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result

arr = list(map(int, input("Enter array: ").split()))
sorted_arr = merge_sort(arr)
print("Sorted array:", sorted_arr)
```

Output

Enter array: 4 8 6 5 9

Sorted array: [4, 5, 6, 8, 9]

=== Code Execution Successful ===